

Andy Rafael Domínguez-Monterroza

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Education

Universidad Complutense de Madrid (UCM), Spain 2021–Present
PhD in Mathematical Engineering, Statistics & Operations Research (expected 2026)
Thesis: Topological and Geometric Approaches to Complex Network Analysis
Supervisors: Prof. Alfonso Mateos-Caballero (UPM), Prof. Antonio Jiménez-Martín (UPM)
Universitat de les Illes Balears, Spain 2016
MSc in Applied Mathematics and Advanced Physics
Universidad Nacional de Colombia 2011
BSc in Mechanical Engineering

Research Interests

Topological and geometric methods in financial and economic time series, networks, and biological signals.
Topological Data Analysis with Machine Learning (persistent homology)
Discrete Differential Geometry (Ricci curvature on graphs, curvature flows)
Random Matrix Theory and spectral analysis of high-dimensional systems
Statistical physics of complex systems (multifractality)

Selected Publications

- Domínguez-Monterroza A., Jiménez-Martín A., Mateos-Caballero A. *Network geometry, topology, and spectral analysis in global stock markets: Insights from using the Ricci Curvature, Euler characteristic, and random matrix theory*. (under review, 2nd round). *PLOS ONE*, Sept 2025 [Preprint](#)
- Domínguez-Monterroza A., Jiménez-Martín A., Mateos-Caballero A. *Correlation structure of the Spanish stock market around COVID-19 using random matrix theory*. *Computational Economics*, 2025. Springer-Nature. DOI:10.1007/s10614-024-10820-0
- Domínguez-Monterroza A., Mateos-Caballero A., Jiménez-Martín A. *Age-dependent patterns of cardiac complexity unveiled by topological data analysis*. *PLOS ONE*, 20(11): e0337620. DOI:10.1371/journal.pone.0337620
- Domínguez-Monterroza A., Mateos-Caballero A., Jiménez-Martín A. *Machine Learning for Parkinson's Disease Detection using spectral and topological features*. *IEEE Open Journal of Computer Society*, Jan 2026, 1-12. DOI: 10.1109/OJCS.2026.3651318
- Domínguez-Monterroza A. *Multifractal luminance and aesthetic complexity in van Gogh's paintings*. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 1 December 2025; 35 (12): 123140. DOI: 10.1063/5.0307352
- Domínguez-Monterroza A. *Quantifying Uncertainty in Human Performance: A Hierarchical Conformal Machine Learning Framework for Marathon Finish Time Prediction*. (under review) – *IEEE Transactions on Artificial Intelligence*, nov 2025 [Preprint](#)

- Sierra-Porta D., Domínguez-Monterroza A. *Linking cosmic ray intensities to cutoff rigidity through multifractal detrended fluctuation analysis. Physica A: Statistical Mechanics and its Applications, Volu 607, 1 Dec 2022, 128159. DOI:10.1016/j.physa.2022.128159*

Research Experience

PhD Researcher, UCM and UPM

2021–Present

- Developed theoretical and computational methods combining persistent homology, Ricci curvature, and random matrix theory.
- Published in top journals across applied mathematics, physics, and biomedical signal analysis.
- Independent research on visibility graphs, multifractal analysis, nonlinear dynamics, and geometry of images/signals.

Teaching Experience

Assistant Professor and Director of the Data Science Program

Universidad Tecnológica de Bolívar, Cartagena de Indias, Colombia

2019–2024

Courses taught: Linear Algebra, Probability, Inferential Statistics, Time Series Analysis, Machine Learning, Introduction to Data Science.

Lecturer (several institutions in Colombia)

2015–2019

Courses taught: Probability, Stochastic Processes, Differential Calculus, Linear Programming, Precalculus.

Awards and Fellowships

- Banco Santander Ibero-American Scholarship (2016).
- Best Undergraduate Thesis Award, Universidad Nacional de Colombia (2011).

Service

- Peer reviewer for: *Biomedical Signal Processing & Control* (Elsevier), *Chaos Solitons & Fractals* (Elsevier), *Computational Economics* (Springer-Nature)
- Co-organizer of the Economics & Data Science Seminar (UTB- Central Bank of Colombia, 2024).

Skills

Python, R, MATLAB; persistent homology (GUDHI, Ripser), Ricci curvature (GraphRicciCurvature), advanced numerical linear algebra.